EV Technology Course — Module 2: Electric Traction Motors

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Most electric cars use interior permanent-magnet synchronous motors (IPMSM) because they are the most efficient and compact, but they depend on rare-earth magnets mostly supplied by China. This module explains how every main motor type works, how to read torque–speed and efficiency maps, how motors are built, and why magnet-free designs are back on the agenda.

1. Learning objectives

- Explain how a rotating magnetic field produces torque.
- Compare IPMSM, BLDC, induction, wound-rotor (EESM), switched reluctance and axial-flux motors.
- Read a torque–speed curve: constant torque, constant power (field weakening), maximum speed.
- Identify motor losses and how cooling removes them.
- Describe how stators, rotors and hairpin windings are manufactured.
- Explain the rare-earth supply risk and the engineering responses.

2. Fundamentals

An electric motor turns electrical energy into rotation using magnetism. A current-carrying wire in a magnetic field feels a force (the Lorentz force). In a motor:

- The **stator** (stationary outer part) holds copper windings. Feeding them three-phase AC from the inverter creates a magnetic field that **rotates** around the stator.
- The **rotor** (inner rotating part) is dragged around by that rotating field. How it is dragged decides the motor type: by permanent magnets (PMSM), by currents induced in the rotor (induction), by an electromagnet in the rotor (EESM), or by the rotor's shape alone (reluctance motors).
- **Synchronous** motors turn exactly with the field; **asynchronous** (induction) motors lag slightly behind (slip).

3. Motor types

Type	How the rotor makes torque	Efficiency	Power density	Magnets	Cost	Typical use
IPMSM (interior PM)	Magnets buried in rotor + reluctance torque	Highest (peak ~96–97%)	Very high	NdFeB, often with Dy/Tb	Medium–high	Most EV cars
Surface PM / BLDC	Magnets on rotor surface; trapezoidal drive	High	High at low speed	NdFeB or ferrite	Low	E-scooters, e-rickshaws, hub motors
Induction (IM)	Rotating field induces current in rotor bars	High (peak ~93–95%), lower at part load	Medium–high	None	Low–medium	Early Tesla, Audi e-tron, secondary axles
Wound-rotor (EESM)	Electromagnet rotor fed through slip rings or wireless exciter	High; tunable field	Medium–high	None	Medium	BMW Gen5, Renault
Switched reluctance (SRM)	Rotor teeth pulled to energised stator poles	Medium–high	Medium	None	Lowest	Niche, research; torque ripple and noise
PM-assisted synchronous reluctance	Mostly reluctance torque, small magnets	High	High	Small amount	Medium	Tesla Model 3 rear motor (IPM-SynRM type)
Axial-flux PM	Flux runs parallel to shaft; disc rotor(s)	High	Highest torque per kg	NdFeB	High	Performance cars (YASA in Mercedes-AMG)

Torque–speed curve. Up to the **base speed**, the motor can deliver its full torque (limited by current). Above base speed the back-voltage of the motor approaches the battery voltage, so the controller weakens the magnetic field (**field weakening**) and power stays roughly constant while torque falls. At very high speed, mechanical stress on the rotor and iron losses set the limit.

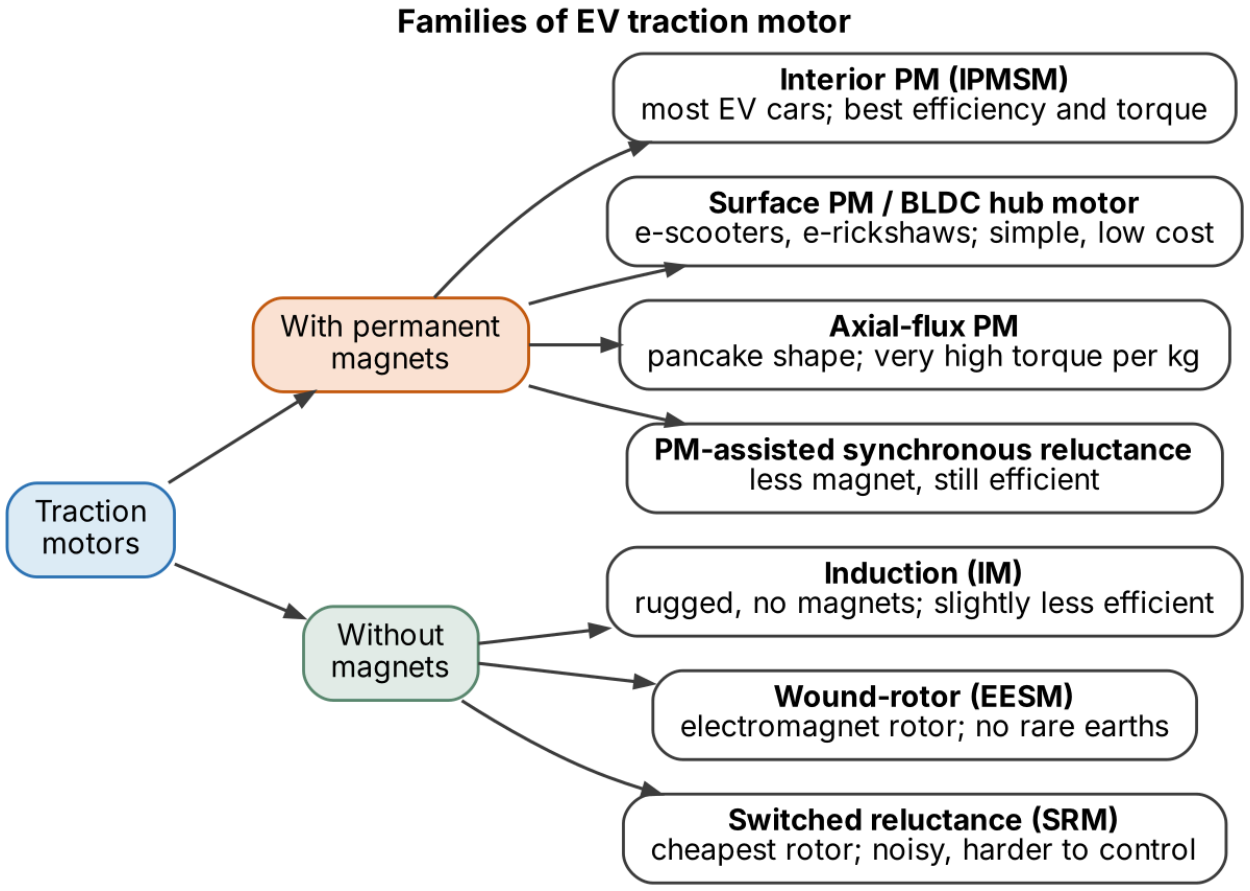
Efficiency map. Efficiency varies with torque and speed. IPMSMs peak around mid speed and moderate torque; induction motors lose more at light load. Real driving spends most time at light load, so part-load efficiency matters more than peak.

Losses and cooling

Loss	Cause	Grows with
Copper (I^2R)	Resistance of windings	Torque (current squared)
Iron (core)	Hysteresis and eddy currents in steel laminations	Speed / frequency
Magnet	Eddy currents in magnets	High frequency
Mechanical	Bearings, windage	Speed

Cooling: water jacket around the stator; oil spray or oil circulating through the shaft and onto end windings in high-performance designs. Magnets lose strength if overheated (above ~150–180 °C for typical grades), which is why heavy rare earths (dysprosium, terbium) are added.

4. Diagram: families of traction motor



5. Step-by-step: how a PMSM makes torque

1. The inverter sends three sine-shaped currents to the stator windings, each shifted by 120°.
2. Together they create a magnetic field that rotates at a speed set by the current frequency.
3. Magnets in the rotor try to line up with the rotating field and are pulled around.
4. The controller keeps the stator field about 90° ahead of the rotor magnets (more in IPMSMs, to add reluctance torque) for maximum torque per amp; it knows where the rotor is from a **resolver** or encoder.
5. More current → more torque. Higher frequency → more speed.
6. Above base speed, the controller injects current that opposes the magnet field (field weakening) so the motor can spin faster without its back-voltage exceeding the battery voltage.
7. On braking, the same machine generates: the rotor drives the field and the inverter rectifies energy back to the battery.

6. Engineering equations

Synchronous speed:

$$n_s = \frac{120 f}{p}$$

Induction-motor slip:

$$s = \frac{n_s - n_r}{n_s}$$

PMSM torque (d–q frame):

$$T = \tfrac{3}{2} p_p \left[\psi_m i_q + (L_d - L_q) i_d i_q \right]$$

Mechanical power and efficiency:

$$P = T \omega = T \frac{2\pi n}{60} \qquad \eta = \frac{P_{out}}{P_{out} + P_{losses}}$$

Copper loss (three-phase):

$$P_{Cu} = 3 I_{ph}^2 R_{ph}$$

n in rpm, f in Hz, p = number of poles, p_p = pole pairs, ψ_m = magnet flux linkage (Wb), i_d and i_q = currents along and across the magnet axis (A), L_d and L_q = inductances (H), I_{ph} = phase current (A rms), R_{ph} = phase resistance (Ω).

7. Worked examples

Example 1: electrical frequency. An 8-pole PMSM at 12,000 rpm: f = 12,000 × 8 ÷ 120 = **800 Hz**. The inverter must switch fast enough to synthesise this cleanly (typically 10–20 kHz switching).

Example 2: power from torque. 300 N·m at 4,000 rpm: P = 300 × 2π × 4,000 ÷ 60 = **125.7 kW**.

Example 3: base speed. Peak 150 kW, peak torque 310 N·m: ω_{base} = 150,000 ÷ 310 = 484 rad/s → **4,621 rpm**. Above this speed, torque falls as power stays near 150 kW.

Example 4: copper loss. Phase current 300 A rms, phase resistance 10 mΩ: $P_{Cu} = 3 \times 300^2 \times 0.010$ = **2.7 kW** of heat in the windings.

Example 5: induction slip. 4-pole motor at 200 Hz: n_s = 6,000 rpm; rotor at 5,880 rpm → s = **2%**.

8. Real-world EV examples

- **Tesla:** original Model S used induction motors; Model 3 rear motor uses a permanent-magnet switched-reluctance design (an IPM-SynRM type); dual-motor versions pair it with an induction front motor that can idle efficiently.
- **BMW Gen5 (iX, i4):** wound-rotor synchronous motor (EESM), no rare earths.
- **Mercedes-AMG:** axial-flux motors from YASA (acquired by Mercedes in 2021) for performance models.
- **Indian cars (Nexon EV, BE 6, Creta Electric):** permanent-magnet synchronous motors.
- **Indian two- and three-wheelers:** BLDC hub motors in many low-cost scooters and e-rickshaws; frame-mounted PMSMs in premium scooters.

9. Manufacturing applications

- **Laminations:** electrical steel 0.2–0.35 mm thick, stamped at high speed (or laser-cut for prototypes), stacked and bonded, welded or interlocked. Thinner steel cuts iron loss but costs more.
- **Windings:** round wire (cheaper, ~40–50% slot fill) or **hairpin** rectangular copper bars (~65–75% slot fill, better cooling, automated, but needs precise forming, twisting and laser welding of hundreds of joints).

- **Impregnation:** varnish or resin trickle impregnation for insulation and heat transfer.
- **Rotor:** magnets inserted into slots and glued or potted; rotor balanced to tight limits for 15,000–20,000 rpm.
- **End-of-line tests:** back-EMF, resistance, insulation and partial-discharge tests (important with fast-switching SiC inverters), noise and vibration.
- **Costing:** magnets (NdFeB with Dy/Tb), copper and electrical steel are the main material cost drivers; hairpin lines are capital-intensive.

10. Common mistakes and misconceptions

- **"BLDC and PMSM are completely different."** Both use magnets; they differ mainly in back-EMF shape and drive (trapezoidal vs sinusoidal).
- **"Peak efficiency decides real efficiency."** Drive-cycle efficiency depends on the part-load region.
- **"Induction motors are obsolete."** They remain useful where magnet cost or drag matters (secondary axles).
- **"Rare-earth-free means worse."** EESMs approach PMSM efficiency, especially at high speed, with different trade-offs.

11. Practical case study: rare-earth magnet supply shock

Problem. From 4 April 2025, China required export licences for rare-earth magnets and related elements. Indian automakers reported licence delays disrupting EV production schedules through mid-2025; Bajaj Auto later received cleared shipments after weeks of disruption, and Mahindra used engineering workarounds and explored alternatives (Outlook Business, 11 Sep 2025).

Engineering responses. Reduce or eliminate heavy rare earths (grain-boundary diffusion of Dy), redesign with ferrite magnets or PM-assisted reluctance, adopt EESM or induction motors, dual-source magnets.

Policy response. On 26 November 2025, India's Cabinet approved a ₹7,280 crore scheme to build 6,000 tonnes per year of sintered rare-earth permanent magnet capacity, through five beneficiaries of up to 1,200 tonnes each, over seven years (Tribune India).

Lesson. Motor choice is a supply-chain decision as much as a technical one.

12. Revision notes

- Stator makes a rotating field; the rotor follows it.
- IPMSM: best efficiency and density; depends on rare earths.

- Induction, EESM and SRM avoid magnets with trade-offs.
- Constant torque to base speed; constant power above it via field weakening.
- $n_s = 120f \div p$; $P = T\omega$; copper loss = $3I^2R$.

13. Glossary

Term	Meaning
Stator / rotor	Stationary windings / rotating part
Back-EMF	Voltage generated by the spinning motor, opposing the supply
Field weakening	Reducing effective magnet field to reach higher speed
Resolver	Rugged rotor-position sensor
Hairpin winding	Rectangular copper bars inserted and welded
NdFeB	Neodymium-iron-boron magnet
Dy / Tb	Dysprosium / terbium: heavy rare earths for heat resistance
Slot fill	Share of the stator slot occupied by copper

14. Knowledge check

Conceptual

1. Why do most EV cars use IPMSMs?
2. What limits torque below base speed and power above it?
3. Why is part-load efficiency more important than peak efficiency?
4. Why do hairpin windings improve power density?
5. What are the trade-offs of switching from IPMSM to EESM?

Numerical

1. A 6-pole motor is fed at 400 Hz. What is its synchronous speed?
2. What power does a motor deliver at 200 N·m and 6,000 rpm?
3. At 100 kW output, losses are copper 3 kW, iron 2 kW, mechanical 0.5 kW. What is the efficiency?

Application

1. An Indian scooter maker faces magnet shortages. Outline three design options and their cost impact.
2. How would you build a should-cost model for a 150 kW IPMSM?

15. Answer key

Conceptual

1. Highest efficiency over the drive cycle and highest torque and power per kg, giving more range and a smaller motor.
2. Below base speed: current limit (heating, inverter rating). Above base speed: voltage limit (back-EMF vs battery voltage), so field weakening keeps power roughly constant.
3. Most driving uses a small fraction of peak power; losses at those operating points determine Wh/km.
4. More copper in the same slot (higher slot fill) and better heat transfer, so more current and torque from the same size.
5. No rare earths and adjustable field (good high-speed efficiency); but rotor copper losses, excitation hardware (brushes/slip rings or wireless exciter) and somewhat lower power density.

Numerical

1. $n_s = 120 \times 400 \div 6 =$ **8,000 rpm**.
2. $P = 200 \times 2\pi \times 6{,}000 \div 60 =$ **125.7 kW**.
3. $\eta = 100 \div (100 + 5.5) =$ **94.8%**.

Application

1. (a) Ferrite-magnet BLDC/PMSM: cheap magnets, bigger and heavier motor. (b) Reduced-Dy NdFeB via grain-boundary diffusion: same design, modest cost. (c) Induction or SRM: no magnets, more complex control, possibly lower efficiency and more noise. Compare cost per kW and range impact.
2. Materials: steel laminations (kg × price + scrap), copper (kg), magnets (kg × NdFeB price incl. Dy), shaft, bearings, housing (aluminium casting), resolver, insulation. Conversion: stamping, winding/hairpin, welding, impregnation, magnet insertion, balancing, testing (machine hours × rates). Overheads, scrap, margin; sensitivity to magnet prices.

16. Further learning

- Hendershot & Miller, *Design of Brushless Permanent-Magnet Machines*.
- Pyrhönen et al., *Design of Rotating Electrical Machines* (Wiley).
- NPTEL, "Electric Vehicles" and "Electrical Machines" courses.
- US DOE / Oak Ridge National Laboratory teardown benchmarking reports on EV motors.

Sources

- Outlook Business: China's export curbs continue to disrupt rare-earth magnet supply (11 Sep 2025)
- Tribune India: Cabinet approves ₹7,280 crore rare-earth permanent magnets scheme (26 Nov 2025)

Next: Module 3, power electronics.